Physics IGCSE	Summary formula	Physics grade 11
Content	**Symbols**	**System international unit (SI unit)**
1. Hooke's Law for Simple Harmonic Motion. **2.** Acceleration as a function of position. **3.** Elastic Potential Energy. **4.** Velocity of an object as a function of position. **5.** Period of an object of mass m moving with simple harmonic motion. **6.** Angular frequency ω of the system. **7.** Position, Velocity, and Acceleration as a Function of Time. **8.** Period of a Simple Pendulum. **9.** Frequency, Amplitude, and Wavelength. **10.** The Speed of Waves on String.	1. $F_s = -kx,\ F_{app} = kx$ 2. $a = -\frac{k}{m}x$ 3. $PE_s = \frac{1}{2}kx^2$ 4. $v = \pm\sqrt{\frac{k}{m}\left(A^2 - x^2\right)}$ 5. $T = 2\pi\sqrt{\frac{m}{k}}$ 6. $\omega = 2\pi f = \sqrt{\frac{k}{m}}$ 7. $x = A\cos\left(2\pi ft\right)$ $v = -A\omega\sin\left(2\pi ft\right)$ $a = -A\omega^2\cos\left(2\pi ft\right)$ 8. $T = 2\pi\sqrt{\frac{L}{g}}$ 9. $v = \lambda f$ 10. $v = \sqrt{\frac{F_T}{\mu}}$	Where, F_s : Spring force (N), k : Spring constant (N/m). x : Change in spring position (spring stretch or compression) (m). a is acceleration as function of position $\left(m/s^2\right)$ k is the spring constant or factor $\left(N/m\right)$. m is the mass of an object $\left(kg\right)$. PE_s is the elastic potential energy (J). v is the velocity as a function of position $\left(m/s\right)$. A is the extreme position or amplitude $\left(m\right)$. T is the period of mass m moving with SHM $\left(s\right)$ ω is angular frequency $\left(rad/s\right)$. f is the frequency $\left(Hz\right)$. L is the length of string $\left(m\right)$. v is the wave speed $\left(m/s\right)$. λ is the wavelength $\left(m\right)$. F is the tension or force on string $\left(N\right)$. $\mu = \frac{m}{L}$ is the mass per unit length $\left(kg/m\right)$. (-) Minus sign indicates that the force always points in the opposite direction to the displacement.
11. Kepler's third law **12.** Newton's law of universal gravitation **13.** Period of a Planet Orbiting the Sun **14.** Period of a Satellite Orbiting Earth **15.** Speed of a Satellite Orbiting Earth at altitude h from the surface of the Earth **16.** The acceleration of objects due to Earth's gravity **17.** Any object with mass is surrounded by a gravitational field **18.** Inertial mass and gravitational mass	11. $\left(\frac{T_A}{T_B}\right)^2 = \left(\frac{r_A}{r_B}\right)^3$ 12. $F = G\frac{m_1 m_2}{r^2}$ 13. $T = 2\pi\sqrt{\frac{r^3}{Gm_S}}$ 14. $v = \sqrt{\frac{Gm_E}{r}}$ 15. $v = \sqrt{\frac{Gm_E}{r+h}}$ 16. $a = g\left(\frac{r_E}{r}\right)^2$ 17. $g = \frac{GM}{r^2}, I = \frac{GM}{r^2}$ 18. $m_{initail} = \frac{F_{net}}{a},\ m_{grav} = \frac{r^2 F_{grav}}{Gm}$	T_A, T_B are the period of planet A and B (s). r_A, r_B are the distance of each planet A and B to the focus (Sun) (m). m_1, m_2 are the masses of two objects (kg). $G = 6.67\times10^{-11} N.m^2/kg^2$ is universal gravitational constant. r is the distance between centers of two objects measured in (m). T is the period of satellite orbiting (s). m_S is the mass of satellite (kg). v is the Speed of a Satellite Orbiting Earth at altitude h from the surface of the Earth (m/s). h is the altitude of satellite orbiting (m). m_E is the mass of earth (kg). a is the acceleration of objects due to Earth's gravity (m/s^2). r_E is the radius of earth (m). $I = g$ is the gravitational field (N/kg). M is the mass of any objects (kg).

REVIEW FOR FINAL EXAM GRADE 11

A. Multiple choice questions

1. Which of the following do waves transfer?
a. Matter **b.** Matter and energy
c. Energy **d.** The medium

2. What is the formula for calculating wave speed?
a. $v = \lambda f$ **b.** $v = \lambda / f$
c. $v = \lambda - f$ **d.** $v = \lambda + f$

3. When a compressional wave travels through a medium, which does matter in the medium move?
a. Backward
b. Forward
c. Perpendicular to the rest position.
d. Along the same direction the wave travels.

4. What is the highest point of a transverse wave called?
a. Crest **b.** Wavelength
c. Compression **d.** Trough

5. If the frequency of the waves produced by a vibrating object increases, how does the wavelength of the wave produced change?
a. It stays the same. **b.** It vibrates.
c. It decreases. **d.** It increases.

6. If the amplitude of a wave changes, which of the following changes?
a. Wave energy **b.** Wave speed
c. Frequency **d.** Refraction

7. Base on the data in the table within textbook, which of the following would be the speed of sound in air at 30 ^{0}C?
a. 340.4 m/s **b.** 353.4 m/s
c. 346.4 m/s **d.** 349.4 m/s

8. A wave has a frequency of 900 Hz and a wavelength of 200 m. At what speed is this wave traveling?
a. 18×10^4 m/s **b.** 4.5 m/s
c. 0.22 m/s **d.** 1800 m/s

9. A wave has a wavelength of 125 meters is moving at a speed of 20 m/s. What is its frequency?
a. 0.166 Hz **b.** 6.25 Hz
c. 0.16 Hz **d.** 0.016 Hz

10. Maximum displacement from equilibrium position is
a. frequency **b.** amplitude
c. wavelength **d.** period

11. A simple pendulum has a period of 2.5 s. What is its period if its length is made four times as large?
a. 0.625 s **b.** 5 s
c. 2.5 s **d.** 3.54 s

12. The magnitude of the gravitational force between two masses is F. If the distance between two masses is doubled, the gravitational force between the masses will become:
a. F/4 **b.** F/2
c. 4F **d.** 2F

13. Wavelength is measured in units of:
a. Hz **b.** m
c. Joule **d.** m/s

14. The speed of a wave with wavelength 10m and frequency of 50Hz is:
a. 0.5m/s **b.** 0.5m/s
c. 5m/s **d.** 500m/s

15. A 0.60 kg mass attached to a spring vibrates at a frequency of 3.0 Hz with an amplitude of 0.13 m. What is the velocity of the mass as it passes through the equilibrium position?
a. 1.23 m/s **b.** 2.45 m/s
c. 0.78 m/s **d.** 3.90 m/s

16. When a mass m is hung from the lower end of a spring of negligible mass, an extension x is produced in the spring. The mass is set into vertical oscillations. The time period of
a. $T = 2\pi\sqrt{\frac{x}{mg}}$ **b.** $T = 2\pi\sqrt{\frac{gx}{m}}$
c. $T = 2\pi\sqrt{\frac{x}{g}}$ **d.** $T = 2\pi\sqrt{\frac{x}{2g}}$

17. The gravitational force between two objects depends on which of the following?
a. their masses and their separation. **b.** their speeds.
c. their charge and their separation.
d. their masses and their charges.

18. A mass is attached to a vertical spring and bobs up and down between points A and B. Where is the mass located when its potential energy is a minimum?
a. at either A or B
b. midway between A and B
c. one-fourth of the way between A and B
d. none of the above

19. A body is placed at height 'h' above the surface of Earth. The value of 'g' is given by:
a. $g_h = \frac{m_E}{(R+h)^2}$ **b.** $g_h = \frac{Gm_E}{(R+h)^2}$
c. $g_h = \frac{Gm_E}{R^2 h}$ **d.** $g_h = \frac{2Gm_E}{R^2}$

20. A simple pendulum consists of a mass M attached to a weightless string of length L. For this system, when undergoing small oscillations
a. the frequency is proportional to the amplitude.
b. the period is proportional to the amplitude.
c. the frequency is independent of the length L.
d. the frequency is independent of the mass M.

21. A simple pendulum has a length of 0.760 m, and its bob has a mass of 365 g. It is released from an angle of 12.0° with respect to the vertical. Assuming simple harmonic motion, what is the frequency of the pendulum's vibration?
a. 0.48 Hz **b.** 0.57 Hz
c. 0.69 Hz **d.** 1.31 Hz

22. Which of the following helped calculate Earth's mass?
a. Inverse square law **b.** Cavendish's experiment
c. Kepler's first law **d.** Kepler's third law

23. Which of the following is true according to Kepler's first law?
a. Paths of planets are ellipses with Sun at one focus.
b. Any object with mass has a field around it.
c. There is a force of attraction between two objects.
d. Force between two objects is proportional to their masses.

24. An imaginary line from the Sun to a planet sweeps out equal areas in equal time intervals. This is a statement of:
a. Kepler's first law **b.** Kepler's second law
c. Kepler's third law **d.** Cavendish's experiment

25. The period of a satellite orbiting Earth depends upon _____.
a. the mass of the satellite
b. the speed at which it is launched
c. the value of the acceleration due to gravity
d. the mass of Earth

26. The inertial mass of an object is measured by exerting a force on the object and measuring the object's _____ using an inertial balance.
a. gravitational force **b.** acceleration
c. mass **d.** force

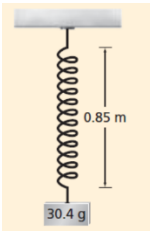
27. Your apparent weight _____ as you move away from Earth's center.
a. decreases **b.** increases
c. becomes zero **d.** does not change

28. A man enters a tall tower and observes a long pendulum hanging from the ceiling to near the floor. He notes that the pendulum has a period of 12.0 s.
(a) What is the approximate height of the tower, assuming the pendulum behaves as a simple pendulum?
a. 14.2 m **b.** 24.5 m
c. 35.8 m **d.** 48.1 m
(b) If the same pendulum is taken to the Moon, where the acceleration due to gravity is 1.67 m/s^2, what would be its new period?
a. 18.7 s **b.** 24.0 s
c. 29.1 s **d.** 33.6 s

29. Two spherical objects, each with a mass of 8.0 kg, are placed a certain distance apart. Use Newton's Law of Universal Gravitation to answer the following:
(a) What is the gravitational force between the two objects if they are 5.0 m apart?
a. 6.67×10^{-9} N **b.** 1.71×10^{-10} N
c. 2.13×10^{-11} N **d.** 4.27×10^{-12} N
(b) What is the gravitational force between them when they are 5.0×10^{1} m apart?
a. 1.71×10^{-12} N **b.** 3.42×10^{-12} N
c. 6.67×10^{-11} N **d.** $8.55 \times 10 - 13$ N

30. What is the value of the spring constant of a spring with a potential energy of 6.67 J when it's stretched 247 mm?
a. 70.2 N/m **b.** 71.1 N/m
c. 142 N/m **d.** 218.7 N/m

31. What is the force acting on a spring with a spring constant of 275 N/m that is stretched 14.3 cm?
a. 2.81 N **b.** 19.2 N
c. 39.3 N **d.** 3.93×10^3 N

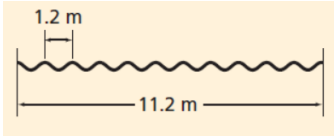
32. A mass stretches a spring as it hangs from the spring. What is the spring constant?


a. 0.25 N/m
b. 0.35 N/m
c. 26 N/m
d. 3.5×10^2 N/m

33. A spring with a spring constant of 350 N/m pulls a door closed. How much work is done as the spring pulls the door a constant velocity from an 85.0-cm stretch to a 5.0-cm stretch?
a. 112 N·m **b.** 126 J
c. 224 N·m **d.** 1.12×10^3 J

34. What is the correct rearrangement of the formula for the period of a pendulum to find the length of the pendulum?
a) $l = \frac{4\pi T^2}{g}$ **b)** $l = \frac{T^2}{4\pi^2}$
c) $l = \frac{T^2 g}{4\pi^2}$ **d)** $l = \frac{Tg}{2\pi}$

35. What is the frequency of a wave with a period of 3 s?
a. 0.3 Hz **b.** 1 Hz
c. 3 Hz **d.** 1/3 Hz

36. A 1.2-m wave travels 11.2 m to a wall and back again in 4 s. What is the frequency of the wave?
a. 0.2 Hz
b. 2 Hz
c. 5 Hz
d. 9 Hz


37. What is the length of a pendulum that has a period of 4.89 s?
a. 5.94 m **b.** 11.9 m
c. 24.0 m **d.** 37.3 m

B. Critical thinking questions

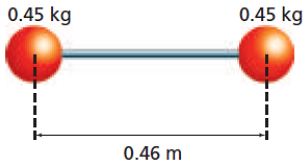
38. What is the Hooke's law for Simple Harmonic Motion?
39. How many types of waves in nature?
40. What is the difference between longitudinal wave and transverse wave?
41. What is the difference between constructive interference and destructive interference? Draw diagram for each phenomenon.
42. What is a Simple Pendulum?
43. What happens to the wavelength of a wave as the frequency increases?
44. What happens to the wavelength of a wave as the frequency increases?

C. Work problems

45. A pendulum with a length of 36.9 cm has a period of 1.22s. What is the acceleration due to gravity at the location of pendulum?

46. If a water wave oscillates up and down three times each second and the distance between wave crests is 2 m, what is its frequency? What is it wavelength? What is its wave speed?

47. A toy consisting of two balls, each 0.45 kg, at the ends of a 0.46-m-long, thin, lightweight rod is shown in Figure 1. Find the moment of inertia of the toy about the center of the rod.



48. Find the gravitational force of attraction between two metal spheres each of mass 90 kg, if the distance between their centers is 6.67 m. Given: $G = 6.67\times10^{-11}(SI)$. Will the force of attraction be different if the same bodies are taken on the moon, their separations remaining the same?

49. A wave has a wavelength of 125 meters is moving at a speed of 20 m/s. What is its frequency?

50. A 100.0g object is attached to a horizontal spring with a spring constant of 10.0 N/m and released from rest with an amplitude of 50.0 cm. What is the speed of the object when it is 30cm from the equilibrium position if the surface is frictionless?

51. A particle moves in simple harmonic motion, and its displacement at any time t is given by the equation:

$$x = 2\cos\left(0.5\pi t + \frac{\pi}{3}\right)(\text{cm})$$

(a) Determine the amplitude of the oscillation.
(b) Calculate the angular frequency ω, and find the time period T of the motion.
(c) Find the maximum velocity of the particle during the oscillation.
(d) Write an expression for the maximum acceleration of the particle.

52. Two 3-kg lead balls are placed with their centers 7.25 m apart. What gravitational force exists between them?

53. Two balls have their centers 2 m apart. One has a mass of 8 kg . The other has a mass of 6 kg . What is the gravitational force between them?

54. a. What is the gravitational force between two $8kg$ spherical masses that are 5 m apart?

b. What is the gravitational force between them when they are 50 m apart?

55. The Hubble telescope has detected the light being emitted from different regions of galaxy M87. The black circle identifies the center of the galaxy. From the characteristics of this light, astronomers have determined that the orbiting speed is 7.5×10^5 m/s for matter located at a distance of 5.7×10^{17} m from the center. Find the mass M of the object located at the galactic center.

56. Two satellites are in circular orbits around the earth. The orbit for satellite A is at a height of 360 km above the earth's surface, while that for satellite B is at a height of 720 km. Find the orbital speed for each satellite.

57. The moon orbits the earth at a distance of 3.85×10^8 m. Assume that this distance is between the centers of the earth and the moon and that the mass of the earth is 5.98×10^{24} kg. Find the period for the moon's motion around the earth. Express the answer in days and compare it to the length of a month.

58. Galileo measured the orbital sizes of Jupiter's moons using the diameter of Jupiter as a unit of measure. He found that Io, the closest moon to Jupiter, had a period of 1.8 days and was 4.2 units from the center of Jupiter. Callisto, the fourth moon from Jupiter, had a period of 16.7 days.

59. Assume that a satellite orbits Earth 225 km above its surface. Given that the mass of Earth is 5.97×10^{24} kg and the radius of Earth is 6.38×10^6 m, what are the satellite's orbital speed and period?

60. A particle moves in a circular path in the counterclockwise direction with a radius of 3.0 m and a constant angular velocity of 8.00 rad/s. At time $t = 0$, the particle is located at $x = 2.0$ m and is moving to the right.
(a) Determine the x-coordinate of the particle as a function of time.
(b) Find the x-component of the velocity and x-component of the acceleration of the particle at any time.

61. A simple pendulum has a length of 50.0 cm and makes 80.0 complete oscillations in 2.00 min. Find;
a. the period of the pendulum
b. the value of g at the location of the pendulum

62. A spring oriented vertically is attached to a hard horizontal surface as in Figure P13.2. The spring has a force constant of 1.46 kN/m. How much is the spring compressed when an object of mass m = 2.30 kg is placed on top of the spring and the system is at rest?

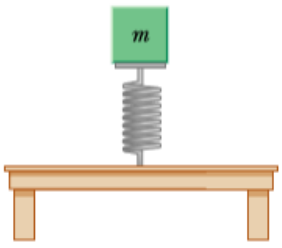


Figure P13.2